

Pattern Level 1 Graduation Review

Rule 1: Never Start Coding First

Always write:

Count = Value = Source =

Examples:

Pattern: 1 12 123

Count = Row Number

Value = Column Number

Source = j

Pattern: 1 22 333

Count = Row Number

Value = Row Number

Source = i

Pattern: 1 23 456

Count = Row Number

Value = Running Counter

Source = Counter Variable

Rule 2: Find the Source

Every pattern value comes from one of:

Constant

i

j

Counter

Formula

Examples:

AAAA

Source = Constant

1234

Source = j

4444

Source = i

123 456 789

Source = Counter

345 4567

Source = Formula

(i + j)

Rule 3: Spaces Are Separate

Never mix stars and spaces.

Make a table.

Row | Spaces | Stars

1 | 4 | 1

2 | 3 | 3

3 | 2 | 5

4 | 1 | 7

5 | 0 | 9

Then derive formulas separately.

Spaces = $n - i$

Stars = $2*i - 1$

Rule 4: Arithmetic Sequence Formula

If values are:

1 3 5 7 9

Start = 1

Difference = 2

Formula:

Start + Difference*(Row-1)

$1 + 2*(i-1)$

$= 2*i - 1$

If values are:

9 7 5 3 1

Start = 9

Difference = -2

Formula:

$9 - 2*(i-1)$

Rule 5: Character Patterns

ASCII:

A = 65

B = 66

C = 67

D = 68

E = 69

Use:

`String.fromCharCode(65)`

to print A

Examples:

ABCDE

Source = j

$65 + j - 1$

A BB CCC

Source = i

$65 + i - 1$

Rule 6: Counter Patterns

Pattern:

1 23 456 78910

Need:

`let counter = 1;`

Counter must be outside the inner loop.

Otherwise it resets every row.

This was one of your biggest mistakes.

Rule 7: Scope Mistake

Wrong:

for(let k...)

use k outside

ERROR

Rule:

Variables declared with let only exist inside their block.

You got stuck on this multiple times.

Always ask:

Does this variable still exist here?

Rule 8: 0-Based vs 1-Based

If:

$i = 0$ to $n-1$

Row number is:

$i + 1$

Many of your formula mistakes came from forgetting this.

Rule 9: Reverse Patterns

Ask:

Start = ?

End = ?

Direction = ?

Example:

54321

Start = 5

End = 1

Direction = Decreasing

Example:

EDCBA

Start = E

End = A

Direction = Decreasing

Rule 10: The Pattern Framework

For every unseen pattern:

1. Count
2. Value
3. Source
4. Spaces
5. Formula
6. Code

Never skip these steps.

This framework is more important than memorizing any pattern.